Monthly Internship Report

Firmware Development Intern at Marshee Pet Tech

Meet Jain

April 12, 2025 – May 17, 2025

1. Introduction

This report outlines my first-month experience as a Firmware Development Intern at Marshee Pet Tech, spanning from April 12, 2025, to May 17, 2025. Marshee is a startup focused on developing advanced BLE-based pet tracking devices with geofencing and cloud integration capabilities. My primary responsibilities involved firmware development on the ESP32-S3 platform, targeting BLE-based communication, OTA updates, and mobile-based geolocation integration.

2. Objectives

The following tasks were officially assigned to me as part of my internship:

1. Implement location services using the GPS of the paired mobile phone.
2. Integrate BLE beacon functionality to ensure compatibility with both Apple (MFi) and Google (Find My Device) ecosystems.
3. Optimize other firmware modules running on the Marshee device for better performance and efficiency.

3. Work Summary

During the internship so far, I have worked extensively on laying the foundation for BLE-based location-aware communication and OTA firmware updates. I began with research on the certification requirements for both Apple's MFi and Find My programs, and Google's Find My Device initiative. This research helped the team understand the compliance needs for integrating BLE beacons with their ecosystem for improved geofencing and tracking.

Next, I developed a GATT (Generic Attribute Profile) server on the ESP32-S3 using the ESP-IDF framework. This GATT server is crucial as it enables the Marshee device to exchange data such as GPS coordinates with a paired mobile phone. To further support BLE-based location and interaction features, I also developed a codebase for Eddystone beacon advertising. This enables the device to broadcast useful location-based data packets that can be recognized by compatible Google and Android devices, aiding in geofencing and device discovery.

To demonstrate the coordinate transmission pipeline, I created a very basic Android application that fetches the mobile's GPS location. Although minimal, the app served as a proof of concept to show how coordinates can later be sent to the ESP32-S3 via BLE using the custom GATT server.

Another major milestone was implementing Over-the-Air (OTA) firmware updates. I successfully created a working system that supports OTA updates using both HTTP and HTTPS, enabling secure and remote updates to the device firmware — a key requirement for any IoT device in the field.

Initially, the firmware development was done in the Arduino IDE. However, recognizing the need for scalability and modularity, I transitioned all development efforts to the ESP-IDF framework. ESP-IDF offers better control, integration, and performance — making it more suitable for professional-grade firmware development.

At present, I am creating a reusable Bluetooth base module using ESP-IDF. This module will act as a foundation for the entire firmware team, allowing them to integrate their logic into a working BLE communication stack for testing and validation. Once this template is complete, I plan to assist the team in migrating their modules into the ESP-IDF environment and help integrate everything into a unified codebase.

4. Tools and Technologies Used

- **Hardware:** ESP32-S3
- **Firmware Platforms:** Arduino IDE (initial phase), ESP-IDF (current)
- **Technologies:** BLE (GATT Server, Beaconing), OTA via HTTP/HTTPS, Mobile GPS services
- **Mobile Development:** Android (for basic coordinate-fetching app)
- **Development Tools:** Git, Visual Studio Code, Android Studio

5. Challenges Faced

Understanding the technical documentation and compliance guidelines for Apple's MFi and Find My, and Google's Find My Device certification was initially challenging due to limited public resources. Additionally, ensuring stable BLE communication while transitioning from Arduino IDE to ESP-IDF required extensive testing. Implementing secure OTA updates via HTTPS also posed some initial configuration and certificate validation hurdles.

6. Key Learnings

Through this phase, I gained hands-on experience with BLE stack development using ESP-IDF, including defining services and characteristics for custom GATT profiles. I learned how OTA updates work under the hood and the necessary steps to ensure secure firmware delivery. The shift to ESP-IDF significantly deepened my understanding of professional embedded systems development practices. Moreover, I acquired a solid foundation on certification procedures for BLE-based consumer devices.

7. Future Work

In the coming weeks, I aim to complete the reusable Bluetooth module and assist the firmware team in testing and integrating their logic into the unified BLE communication framework. I will also continue working on converting other existing modules into the ESP-IDF format. Once all individual modules are integrated and tested, we will move toward preparing for formal certification and field testing.

8. Conclusion

The first month of this internship has been both challenging and rewarding. I have made substantial progress in executing my core responsibilities—implementing location services, BLE beacon integration, and firmware optimization. I am confident that the foundation laid in this phase will contribute significantly to the success of the Marshee pet tracking device. I look forward to furthering my contributions in the next phase of development.

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